Assignment VII

1. **Z-Gate**
   Use hand calculation and co-lab to show that the Z-gate converts $|-\rangle$ to$|+\rangle$.

2. **Controlled-Z Gate**
   Construct a controlled-Z gate from the controlled phase shift gate. Then show that it is a CNOT gate in the $|+\rangle$ / $|-\rangle$ basis.

3. **Implementation of a SWAP gate**
   A SWAP gate can be implemented using 3 CNOT gates in sequence. We will discuss this later. But try it now to challenge yourselves. The first and the third CNOT gates have the first bit as the control bit, but the second CNOT gate has the second qubit as the control bit. Try to create the matrix of the second CNOT gate and then prove the circuit is indeed a SWAP gate.

4. **1-qubit Hadamard Gate**
   Show Equation below:

$$\boldsymbol{H}|\psi\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}1 & 1\\ 1 & -1\end{pmatrix}\begin{pmatrix}\alpha\\ \beta\end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix}\alpha+\beta\\ \alpha-\beta\end{pmatrix}$$
$$= \alpha|+\rangle + \beta|-\rangle$$

   by substituting $|+\rangle$ and $|-\rangle$ in terms of $|0\rangle$ and $|1\rangle$.

5. ***n*-qubit Hadamard Gate Numerical Substitution 1**
   Use Equation below to compute $\boldsymbol{H}^{\otimes 2}|2\rangle$. What is *n* in this case?

$$\boldsymbol{H}^{\otimes n}|y\rangle = \frac{1}{2^{n/2}}\sum_{x=0}^{2^n-1}(-1)^{x.y}|x\rangle$$

6. ***n*-qubit Hadamard Gate Numerical Substitution 2**
   Use expansion method in Equation below

$$\begin{aligned}\boldsymbol{H}^{\otimes n}|y\rangle &= \boldsymbol{H}^{\otimes n}|y_{n-1}y_{n-2}\cdots y_1y_0\rangle\\ &= (\boldsymbol{H}||y_{n-1}\rangle)\otimes(\boldsymbol{H}||y_{n-2}\rangle)\otimes\cdots\otimes(\boldsymbol{H}||y_0\rangle)\\ &= (\tfrac{1}{\sqrt{2}}(|0\rangle+(-1)^{y_{n-1}}|1\rangle))\otimes(\tfrac{1}{\sqrt{2}}(|0\rangle+(-1)^{y_{n-2}}|1\rangle))\\ &\quad\otimes\cdots\otimes\left(\tfrac{1}{\sqrt{2}}(|0\rangle+(-1)^{y_0}|1\rangle)\right)\\ &= \tfrac{1}{2^{n/2}}(|0\rangle+(-1)^{y_{n-1}}|1\rangle)\otimes(|0\rangle+(-1)^{y_{n-2}}|1\rangle)\otimes\cdots\otimes\\ &\quad(|0\rangle+(-1)^{y_0}|1\rangle)\end{aligned}$$

to compute $\boldsymbol{H}^{\otimes 2}|2\rangle$. Try to identify the fact that only when x_i and y_i are both 1 would it get a contribution of −1 to the final coefficient.

7. *n*-qubit Hadamard Gate Numerical Substitutions 3

Compute $\boldsymbol{H}^{\otimes 2}|2\rangle$ but use matrix multiplications.

8. SWAP Gate Implementation

We may change the implementation of the SWAP gate using LSB-controlling CNOT gate followed by MSB-controlling CNOT gate and then followed by LSBcontrolling CNOT gate. Please repeat what we have done in the main text and prove that using the intuition method and matrix multiplication.

9. LSB-Controlling CNOT Gate

Repeat the definition and derivation of the CNOT gate matrix in Chap.14 to obtain the matrix of LSB-controlling CNOT gate. That is to derive $U_{XOR,center}$ in the following Eq.

$$U_{XOR,right}U_{XOR,center}U_{XOR,left} = \begin{pmatrix}1&0&0&0\\0&1&0&0\\0&0&0&1\\0&0&1&0\end{pmatrix}\begin{pmatrix}1&0&0&0\\0&0&0&1\\0&0&1&0\\0&1&0&0\end{pmatrix}\begin{pmatrix}1&0&0&0\\0&1&0&0\\0&0&0&1\\0&0&1&0\end{pmatrix}$$
$$= \begin{pmatrix}1&0&0&0\\0&0&0&1\\0&1&0&0\\0&0&1&0\end{pmatrix}\begin{pmatrix}1&0&0&0\\0&1&0&0\\0&0&0&1\\0&0&1&0\end{pmatrix} = \begin{pmatrix}1&0&0&0\\0&0&1&0\\0&1&0&0\\0&0&0&1\end{pmatrix}$$
$$= U_{SWAP}$$

10. Quantum Computing Exercise 1

Run the basis transformation circuit in Fig. 1 on IBM-Q using real hardware and simulation. Do necessary modification to change the input to |11⟩ and check the output.

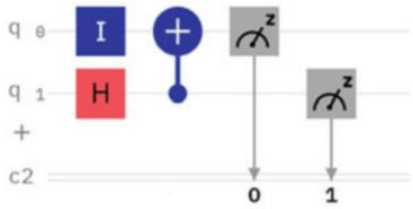


Figure 1 The Quantum Circuit

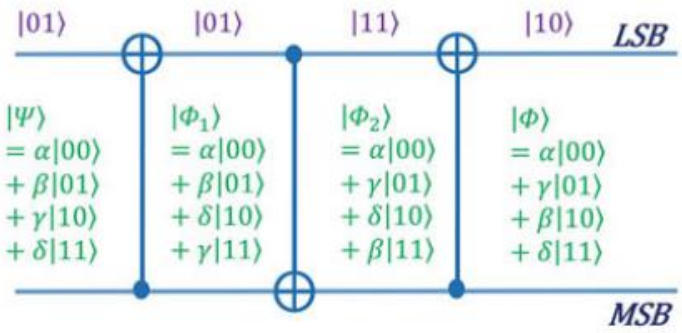


Figure 2 Implementation of a SWAP gate using CNOT gates. The purple ket's on top shows an example of how a single basis vector evolves in the circuit.

11. Quantum Computing Exercise 2

Run the SWAP circuit in Fig. 2, on IBM-Q using real hardware and simulation. Do necessary modification to change the input to |10⟩ and check the output.

12. Quantum Computing Exercise 3

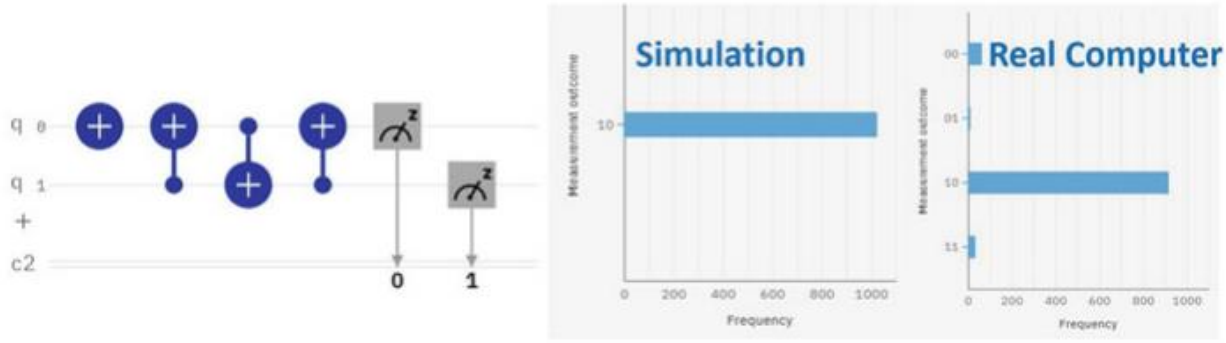


Figure 3 The quantum circuit in Fig. 1 built in IBM-Q. The simulation result is shown in the middle, and the real quantum computer execution result is shown on the right.

Run the SWAP circuit in Fig. 2 on IBM-Q using real hardware and simulation but use the SWAP gate directly instead of using CNOT gates. Compare the outputs to Fig. 3.